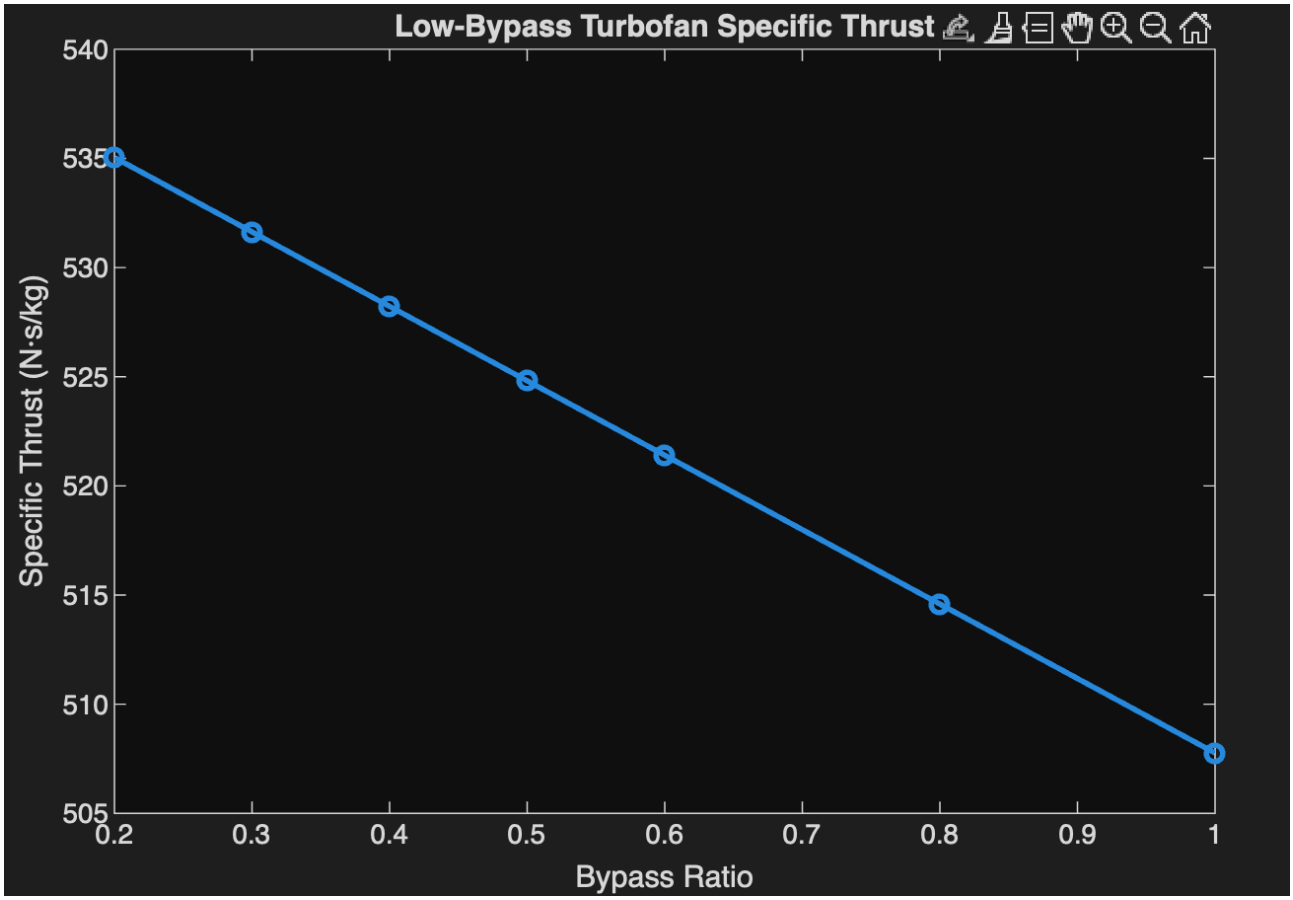
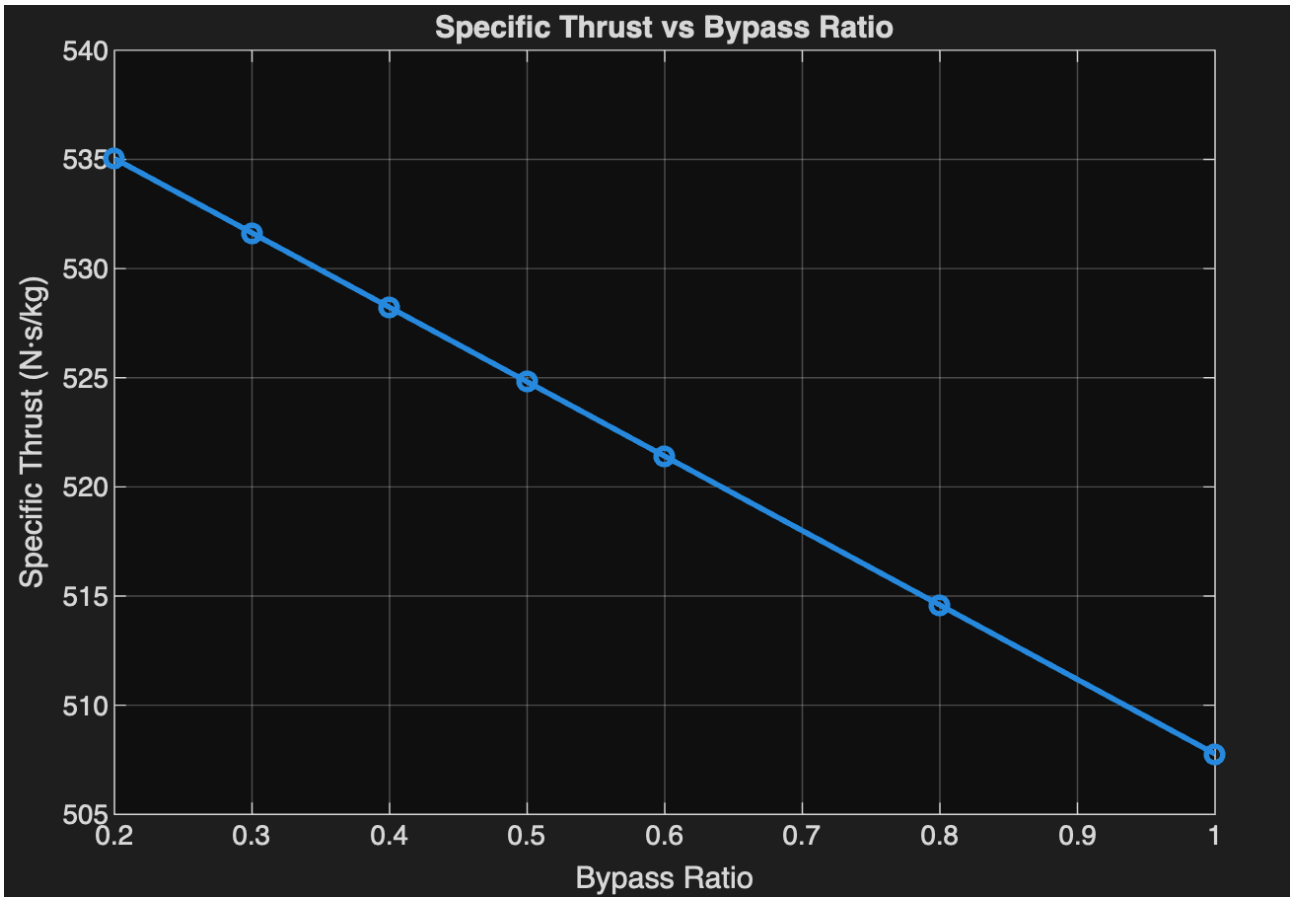
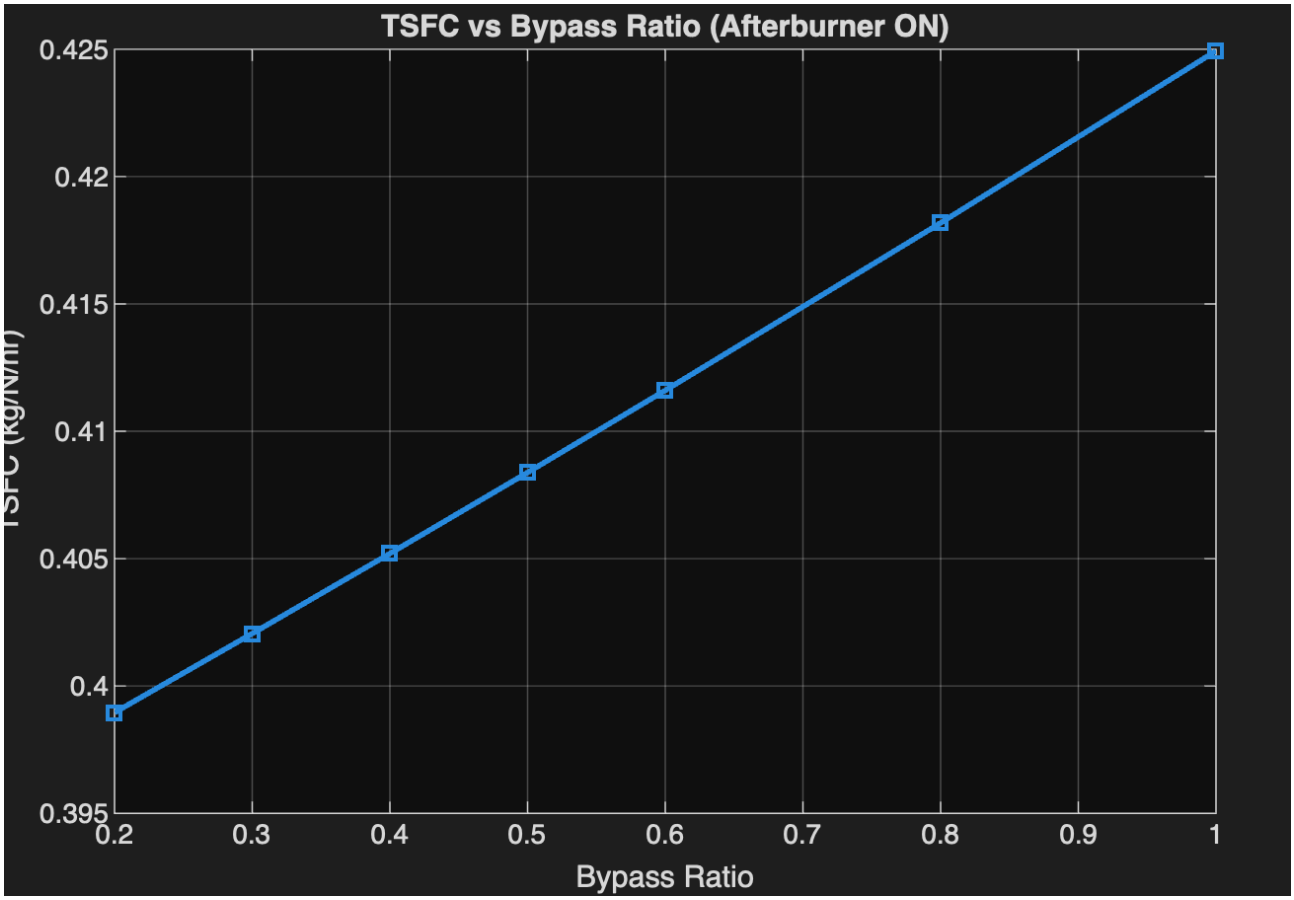
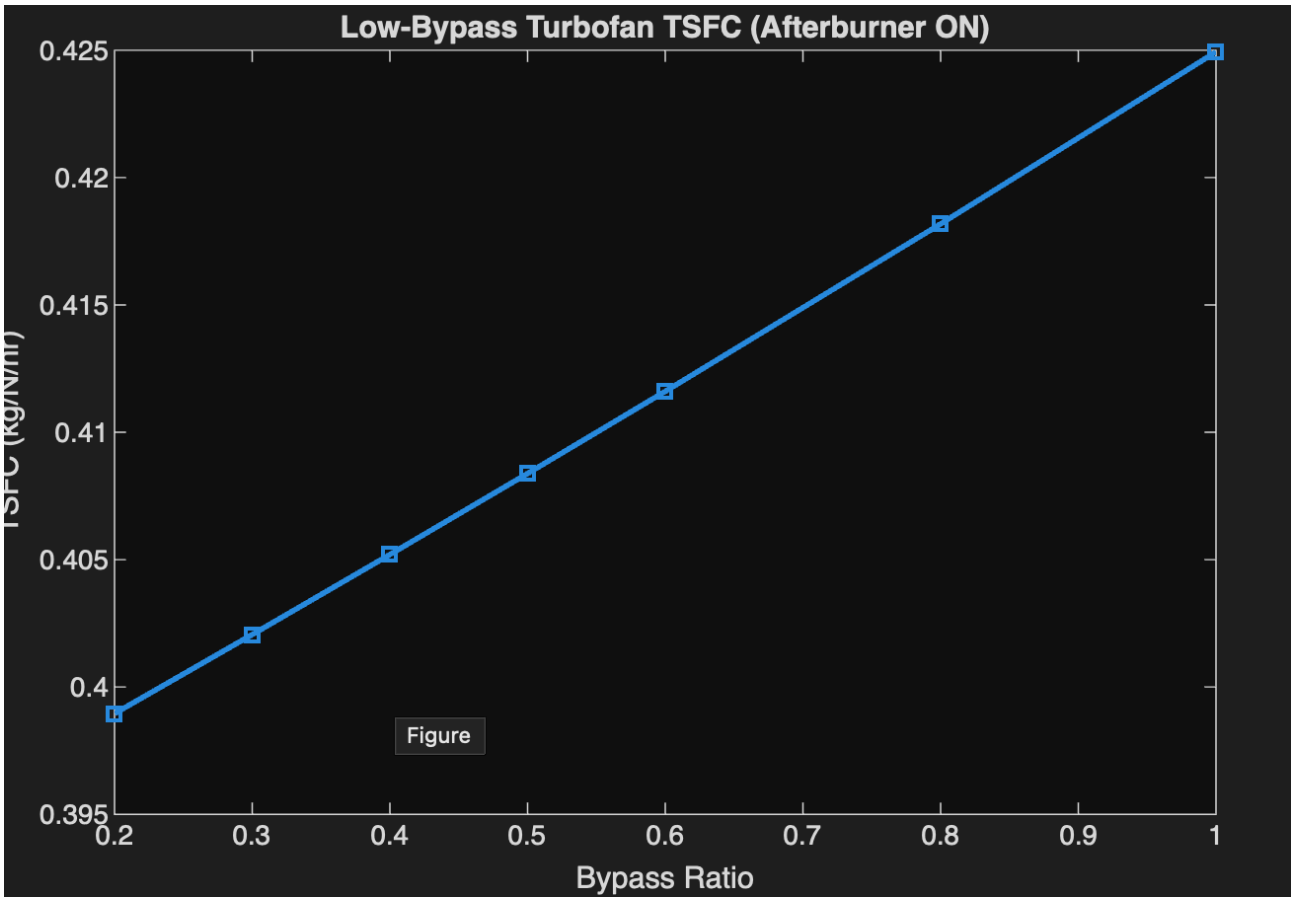
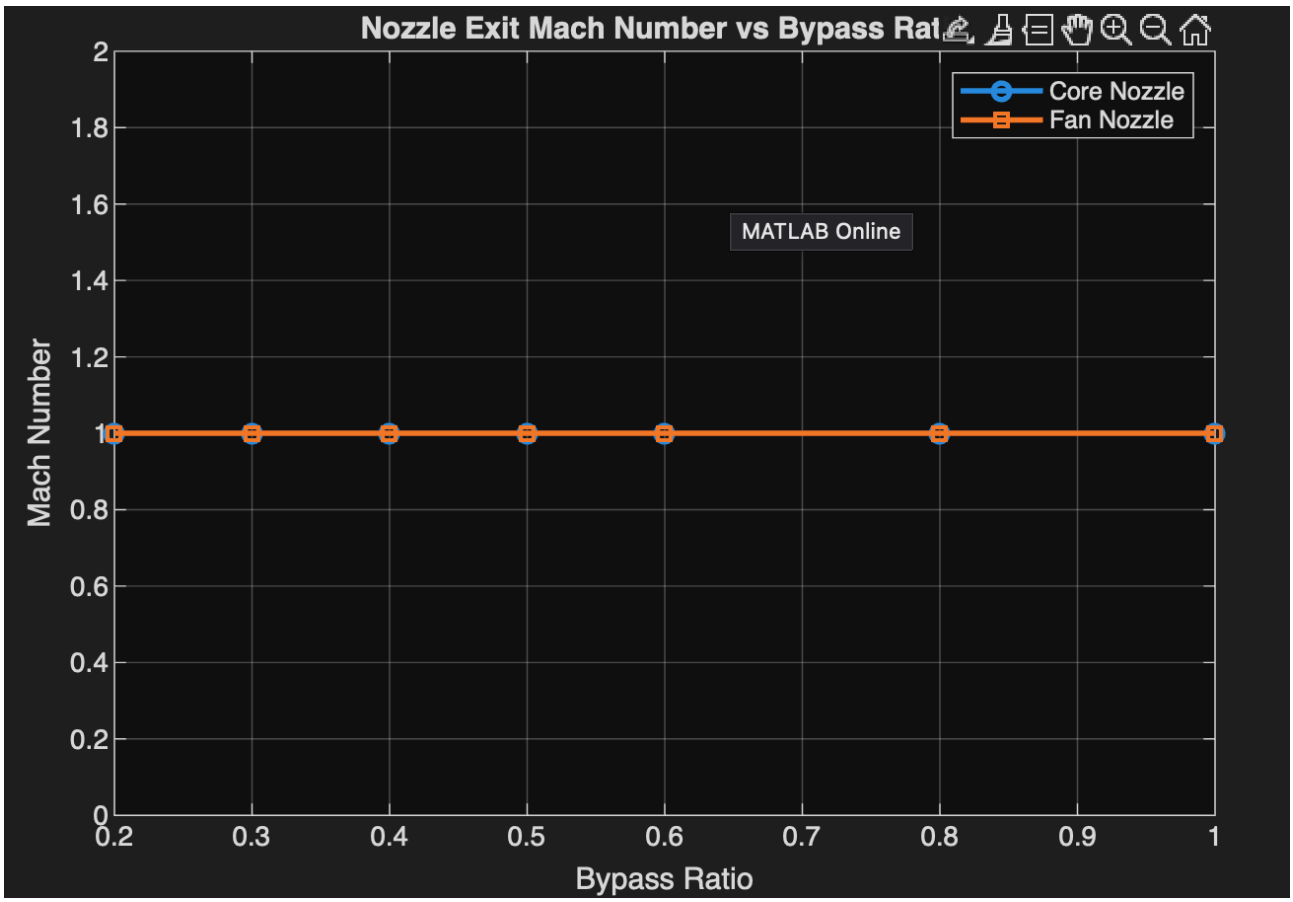
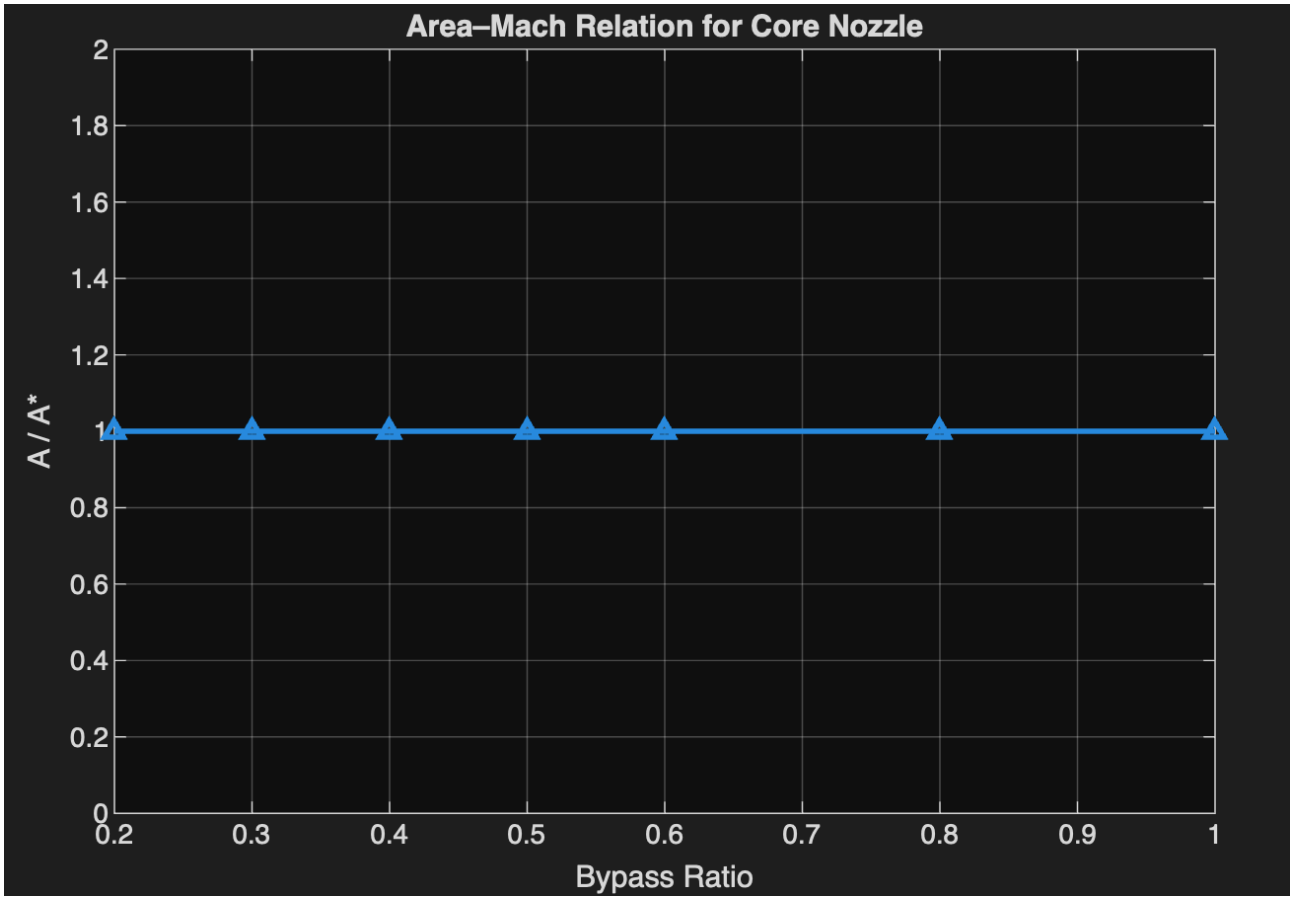

Low-Bypass Turbofan Specific Thrust
Specific Thrust (N·s/kg)
540
535
530
525
520
515
510
505
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

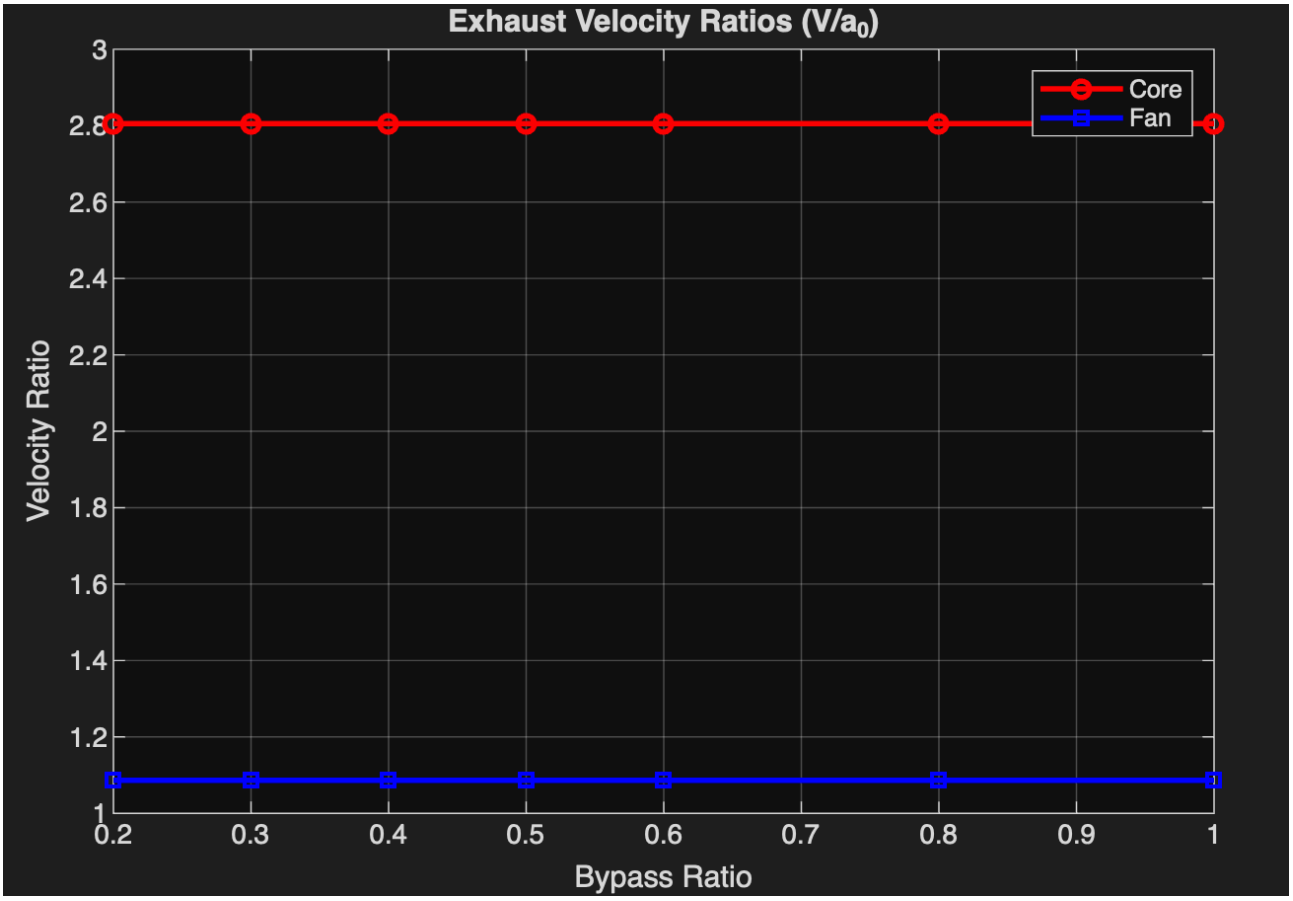

Specific Thrust vs Bypass Ratio
Specific Thrust (N·s/kg)
540
535
530
525
520
515
510
505
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

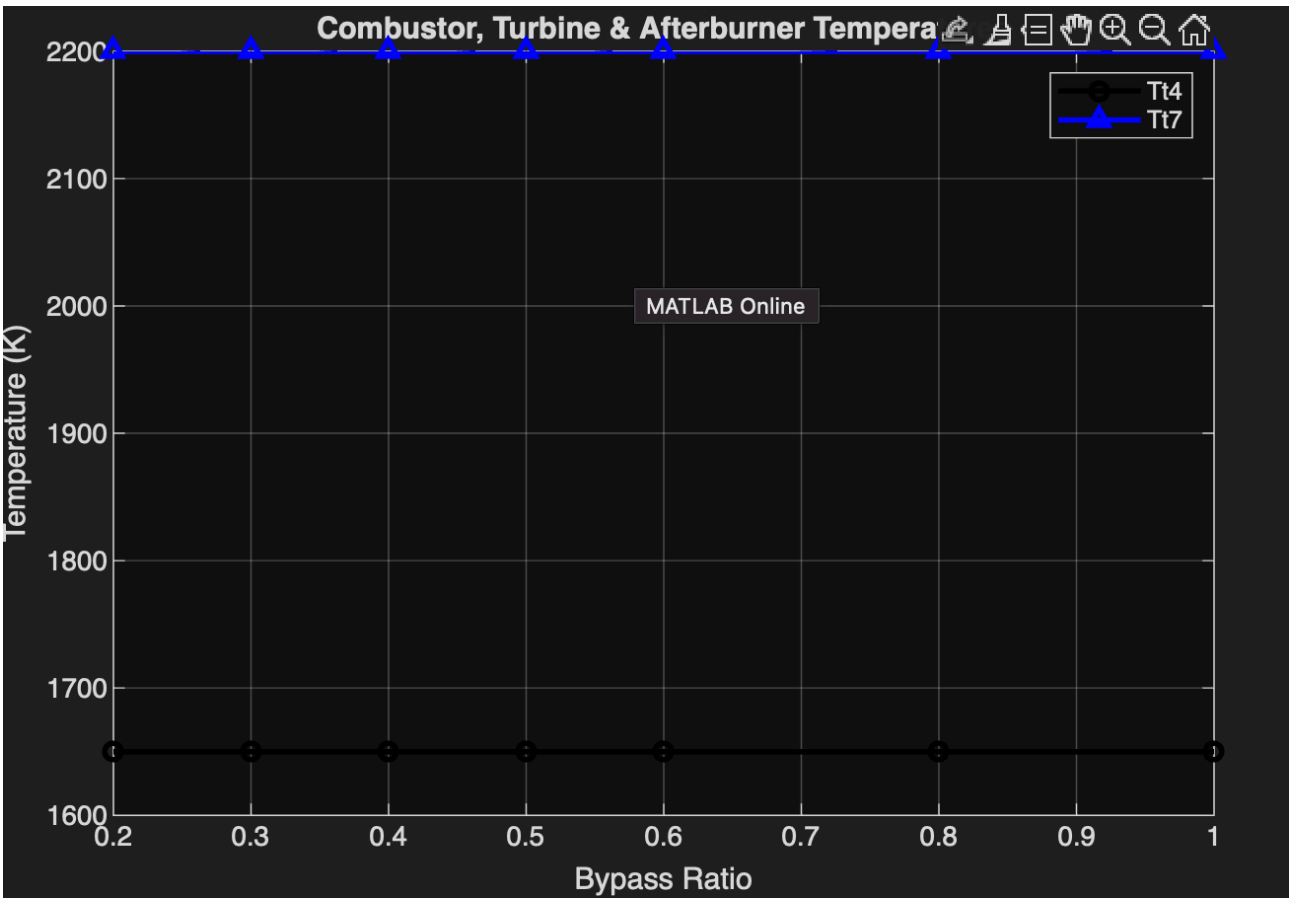

TSFC vs Bypass Ratio (Afterburner ON)
0.425
0.42
0.415
0.41
0.405
0.4
0.395
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio

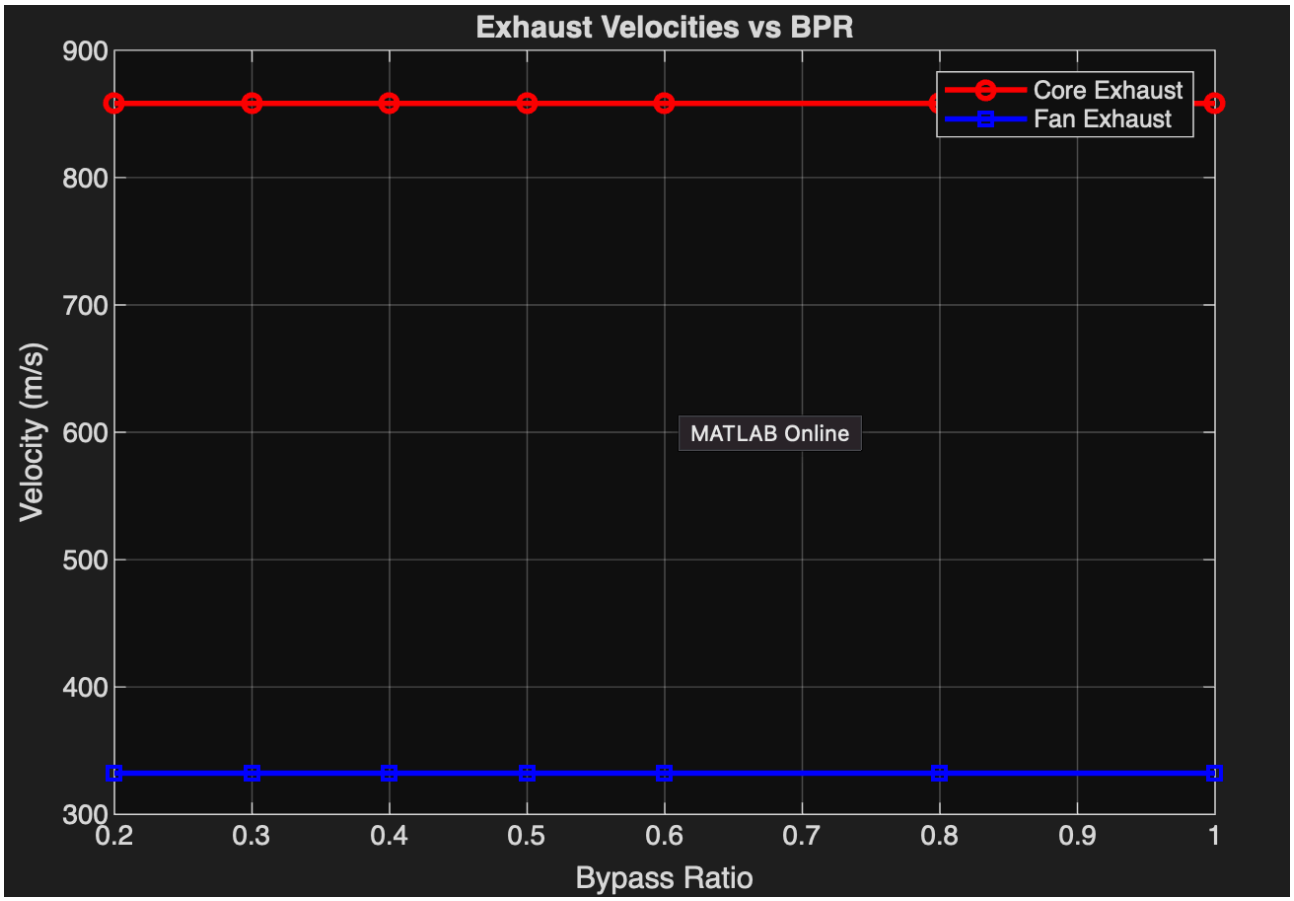

Low-Bypass Turbofan TSFC (Afterburner ON)
0.425
0.42
0.415
0.41
0.405
0.4
0.395
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio
Figure

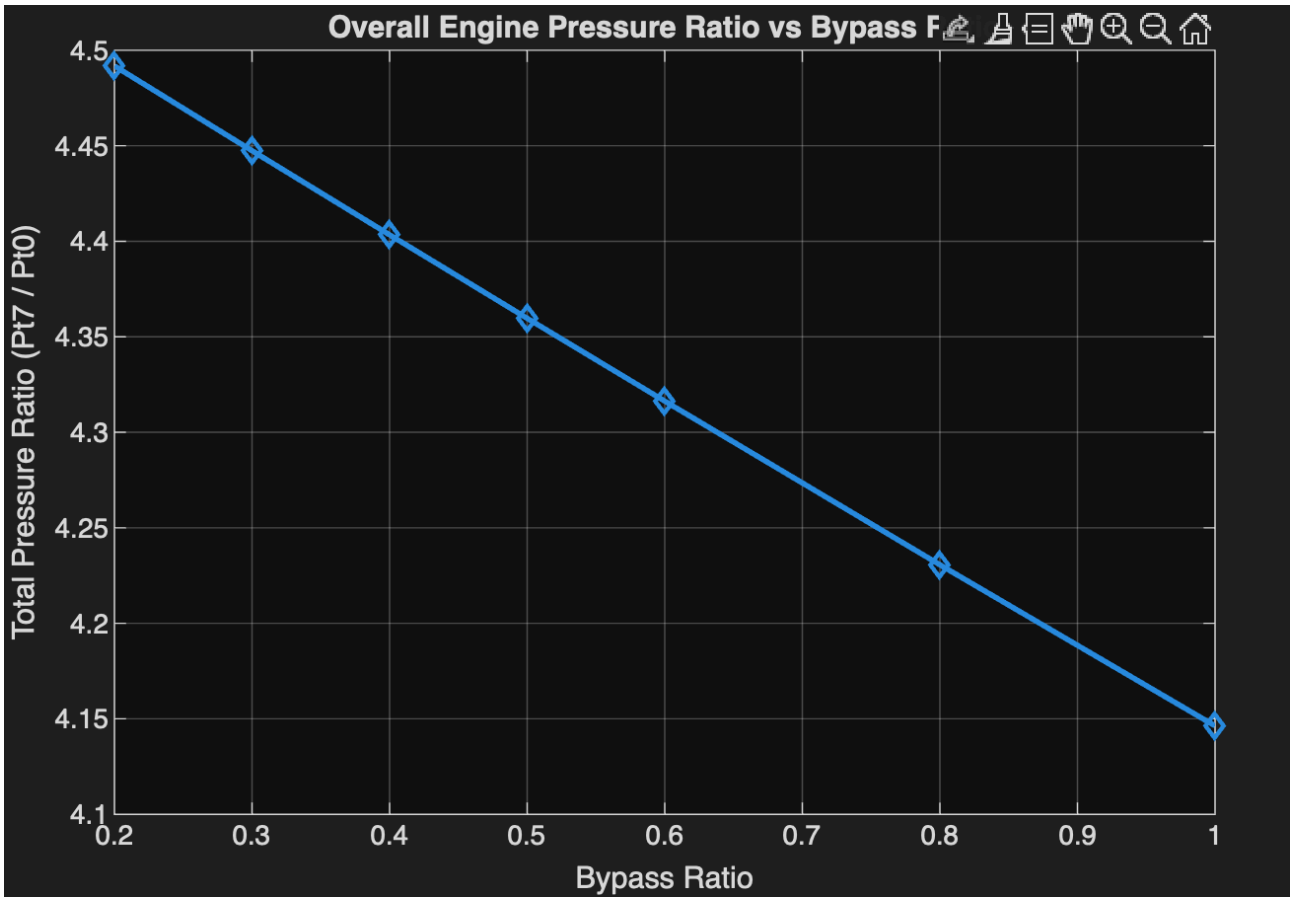

Nozzle Exit Mach Number vs Bypass Rat
Core Nozzle
Fan Nozzle
MATLAB Online
Mach Number
Bypass Ratio
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

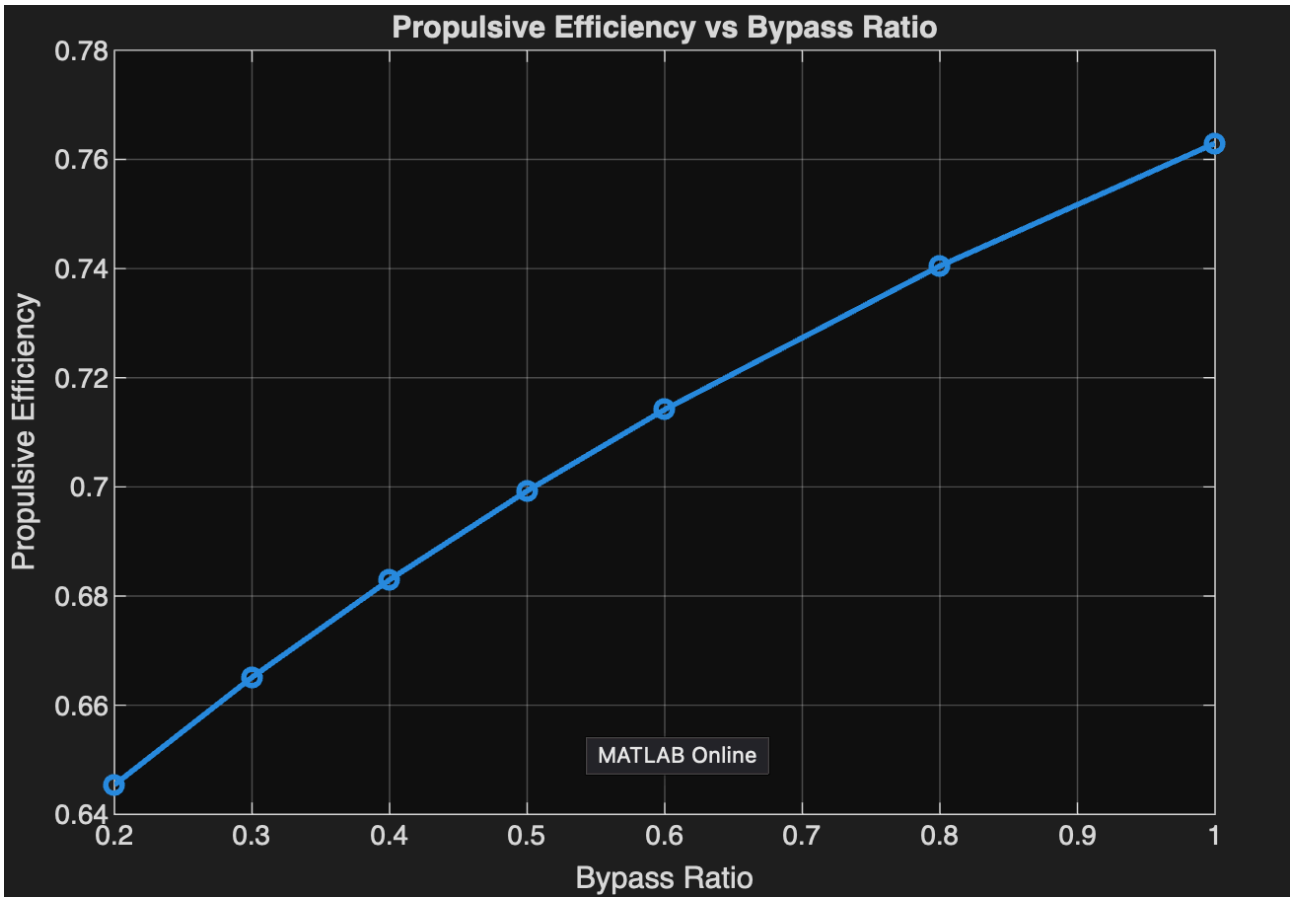

Area–Mach Relation for Core Nozzle
A / A*
Bypass Ratio
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

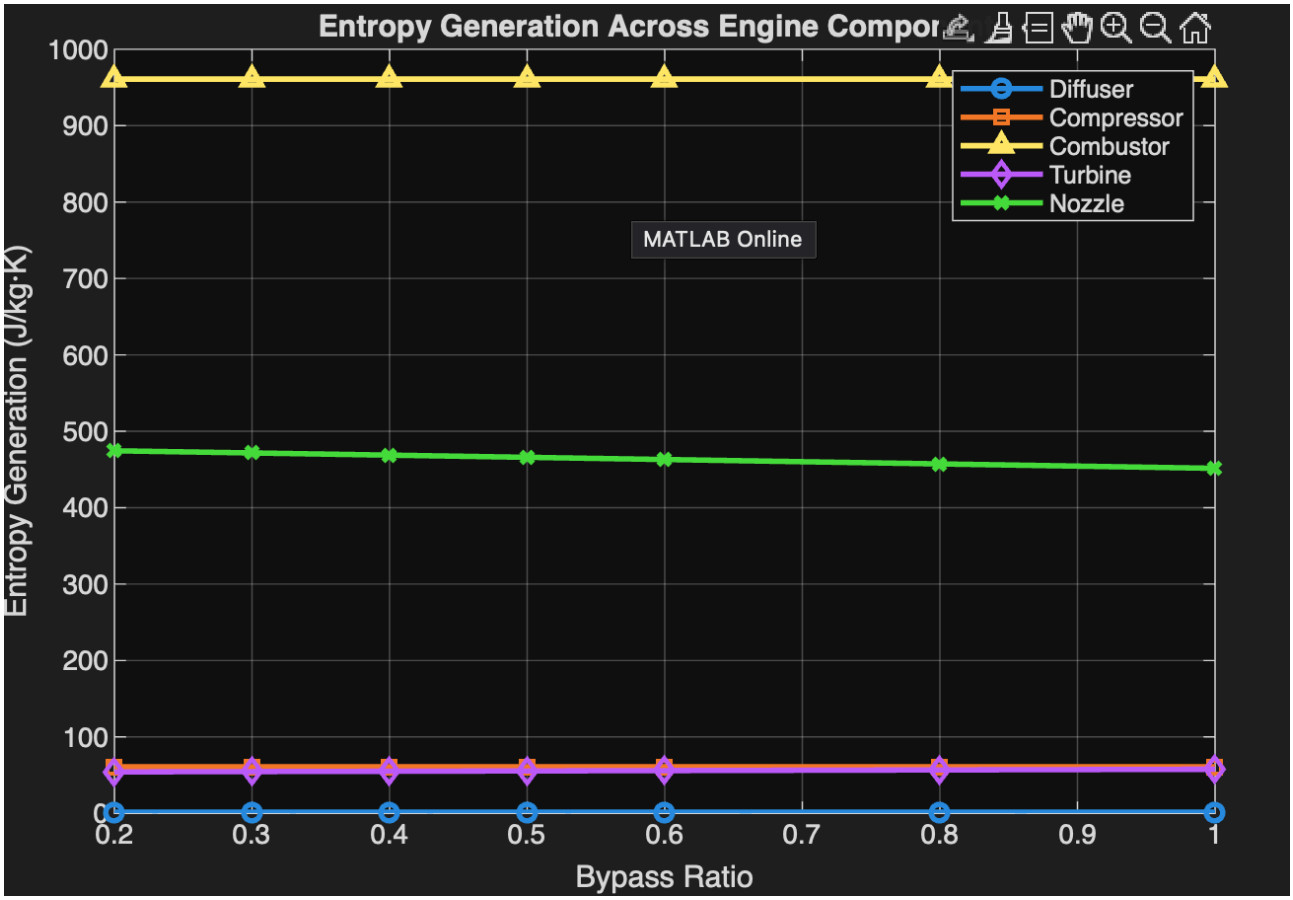

Exhaust Velocity Ratios (V/a0)
Core
Fan
Velocity Ratio
Bypass Ratio
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4
1.2
1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

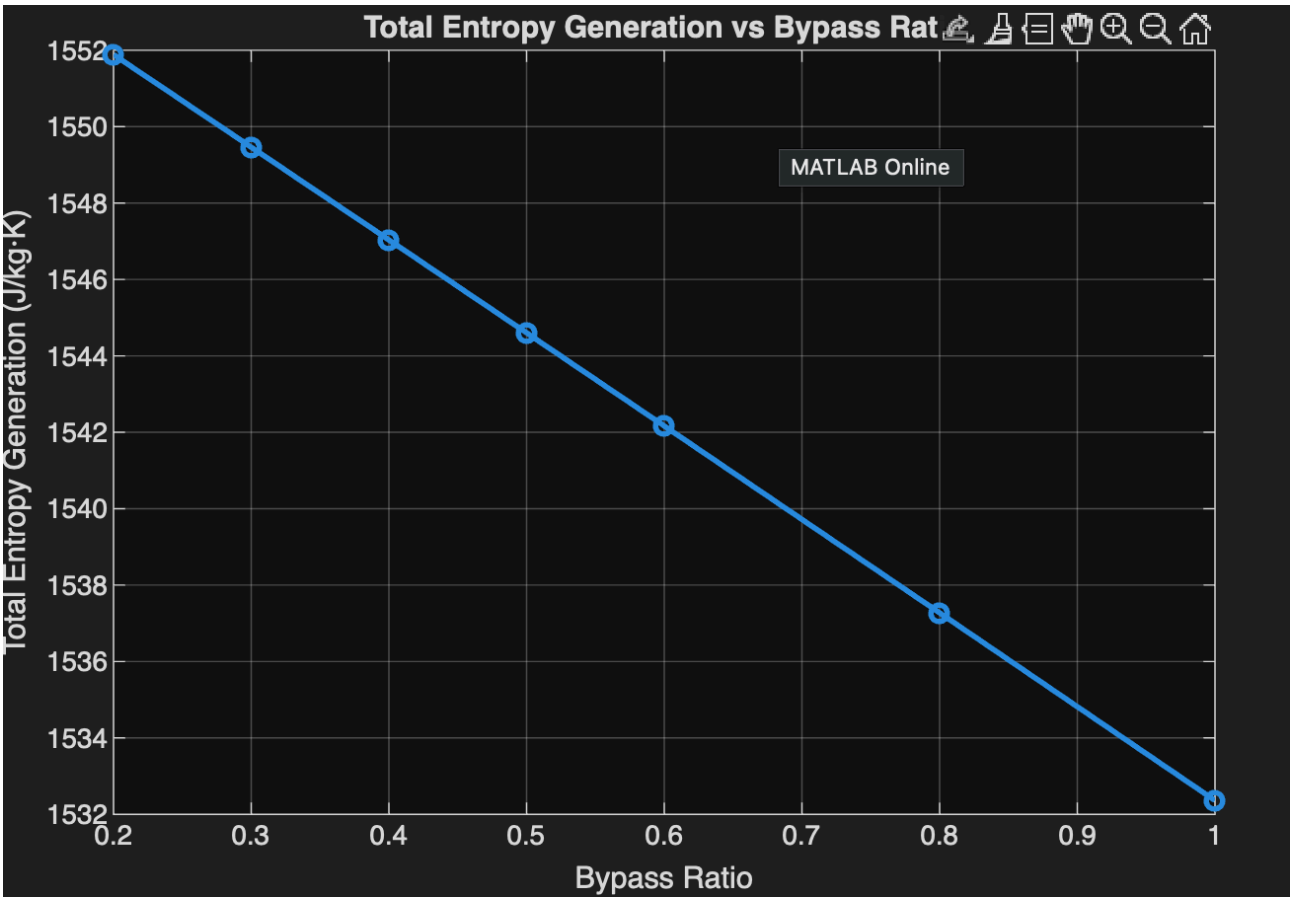

Combustor, Turbine & Afterburner Tempera
Tt4
Tt7
MATLAB Online
Temperature (K)
Bypass Ratio
2200
2100
2000
1900
1800
1700
1600
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

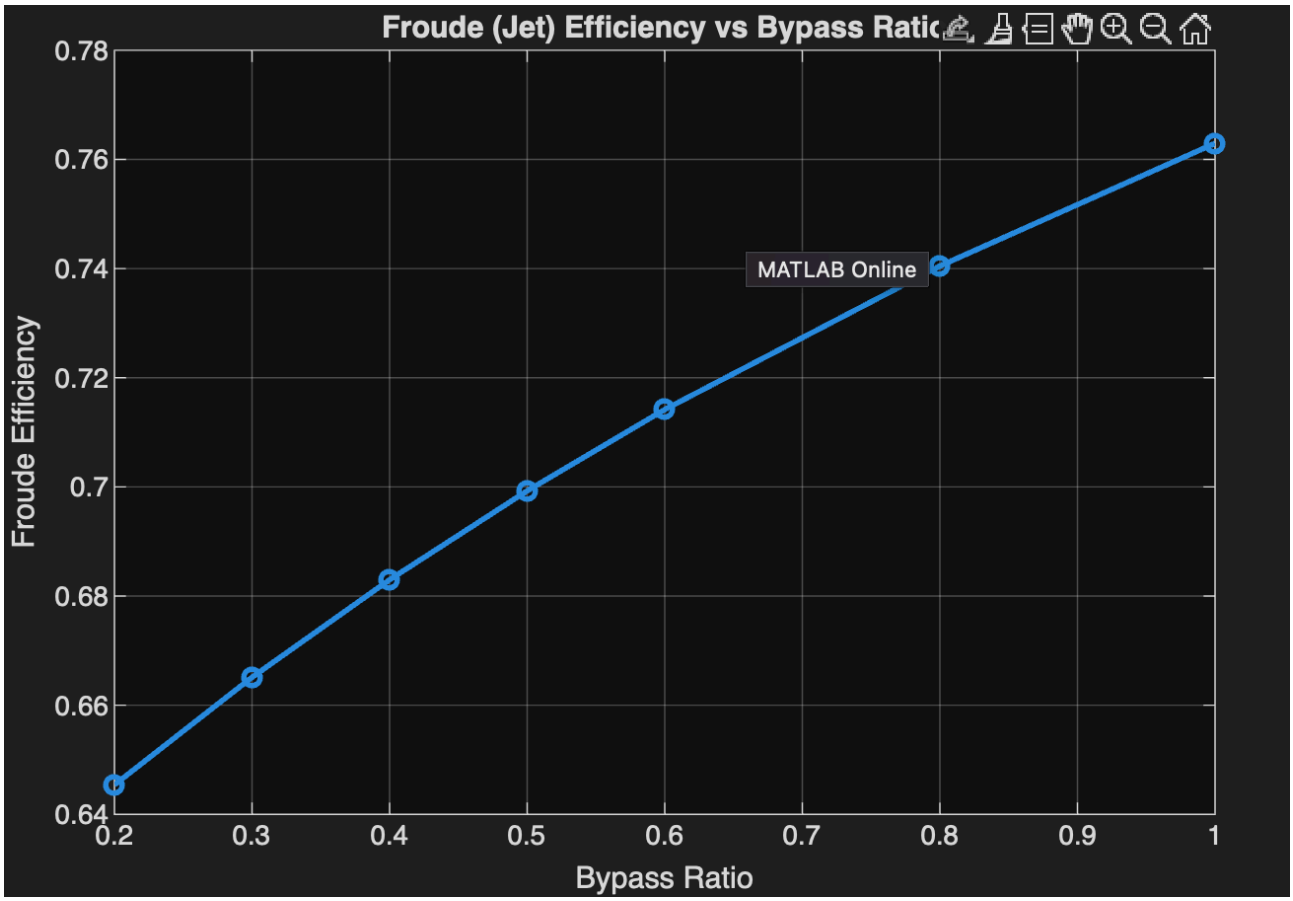

Exhaust Velocities vs BPR
Core Exhaust
Fan Exhaust
MATLAB Online
Velocity (m/s)
900
800
700
600
500
400
300
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Overall Engine Pressure Ratio vs Bypass F
Total Pressure Ratio (Pt7 / Pt0)
4.5
4.45
4.4
4.35
4.3
4.25
4.2
4.15
4.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Propulsive Efficiency vs Bypass Ratio
Propulsive Efficiency
0.78
0.76
0.74
0.72
0.7
0.68
0.66
0.64
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio
MATLAB Online


Entropy Generation Across Engine Compor
Entropy Generation (J/kg·K)
1000
900
800
700
600
500
400
300
200
100
0
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio
Diffuser
Compressor
Combustor
Turbine
Nozzle
MATLAB Online


Total Entropy Generation vs Bypass Rat
MATLAB Online
Total Entropy Generation (J/kg·K)
1552
1550
1548
1546
1544
1542
1540
1538
1536
1534
1532
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio


Froude (Jet) Efficiency vs Bypass Ratio
MATLAB Online
Froude Efficiency
0.78
0.76
0.74
0.72
0.7
0.68
0.66
0.64
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Bypass Ratio